

4 First-order logic – what to read?

Predictably, a *very* large number of texts provide introductions to FOL (in various styles and at various levels of accessibility). Which of them are particularly worth recommending for self-study?

§4.1 warmly recommends four entry-level texts (a gentle introduction, a slightly brisker complementary pair, and a shorter single-text alternative).

§4.2 suggests some alternatives for parallel or slightly more advanced reading.

§4.3 briefly notes some illuminating reading on the history of FOL.

§4.4 is an extended postscript, looking at some of the other literature on FOL, old and new.

A preliminary remark. If you are a philosopher who has worked through a substantial introductory logic text, you could already be familiar with (versions of) a fair amount of the material on FOL. However, in the readings suggested in this chapter, you will now begin to see topics being re-presented in the sort of mathematical style and with the sort of rigorous detail that you will necessarily encounter more and more as you progress in logic. You do need to start feeling entirely comfortable with this mode of presentation. So you will want to revisit even rather familiar topics, this time tackling them with greater mathematical precision.

4.1 Basic recommendations for reading on FOL

The whole point of this Guide is to choose – so let me just dive in without further ado, to give my top recommendations of introductory texts (at three slightly different levels of sophistication). First:

1. Derek Goldrei, *Propositional and Predicate Calculus: A Model of Argument* (Springer, 2005). Read the first five chapters, perhaps skipping §§4.4 and 4.5 for now.

This book is quite explicitly designed for self-study, and a great amount of thought has gone into making this text as clear and helpful as possible. A gently-paced book, this allows Goldrei to be expansive about fundamentals, and to give a lot of examples and exercises with worked answers to test comprehension along the way.

In terms of content, this is an entirely orthodox book (it is chosen for its accessibility, not for presentational novelty). Chapter 2 introduces the syntax and (classical, two-valued) semantics of propositional languages. Chapter 3 explores a Hilbert-style axiomatic system for propositional logic, proving soundness and completeness. Chapter 4 gives a standard account of the syntax and semantics of first-order languages. Chapter 5 gives an axiomatic system for FOL, again proving soundness and completeness. All this is done along conventional lines, but with the greatest clarity: I like the book's tone and approach a great deal.

Note, though, that since Goldrei uses an axiomatic system throughout, you will eventually want to supplement his book with a little reading on a Gentzen-style natural deduction proof system (as in the next recommended book).

Some, however, will perhaps find that Goldrei goes too slowly (though I'd say that this is erring on the right side in an introductory book for self-teaching). Slightly more briskly, then, and together covering both natural deduction and axiomatic proof systems, we have the following admirable pair of texts:

2. Ian Chiswell and Wilfrid Hodges, *Mathematical Logic* (OUP, 2007), up to §7.6.

This is very approachable. It is written by mathematicians primarily for mathematicians, yet it is only one notch up in actual difficulty from some introductory texts for philosophers. However – as its title might suggest – it does have a notably more mathematical ‘look and feel’ (philosophers can cheerfully skip over a few of the more mathematical illustrations).

The headline news is that the authors explore a Gentzen-style natural deduction system. But by building things up in three stages – so after propositional logic, they consider an important fragment of first-order logic before turning to the full-strength version – they make the proof of the completeness theorem for full first-order logic unusually comprehensible. For a more detailed description, see my book note on C&H, tinyurl.com/CHbooknote.

Very warmly recommended. You should then complement C&H by reading the first third of the following excellent book:

3. Christopher Leary and Lars Kristiansen's *A Friendly Introduction to Mathematical Logic** (1st edn by Leary alone, Prentice Hall, 2000; 2nd edn Milne Library, 2015). Downloadable at tinyurl.com/friendlylogic. Chs 1–3, in either edition, do indeed make a particularly friendly and helpful introduction to FOL.

These authors use an axiomatic system, complementing C&H, though this is done in a particularly smooth way. At this stage you could stop reading after

the beginning of §3.3 on compactness, which means you will be reading just 87 pages.

Unusually, L&K dive straight into a treatment of first-order logic without spending an introductory chapter or two on propositional logic: in a sense, as you will see, they let propositional logic look after itself (by just helping themselves to all instances of tautologies as axioms). But this rather happily means (in the present context) that you won't feel that you are labouring through the very beginnings of logic one more time than is really necessary – this book therefore dovetails very nicely with C&H.

Some of L&K's illustrations of ideas can presuppose a smattering of background mathematical knowledge; but philosophers will miss very little if they occasionally have to skip an example (and the curious can always resort to Wikipedia, which is quite reliable in this area, for explanations of some mathematical terms). The book ends with extensive answers to exercises.

I like the overall tone of L&K very much, and say more about this excellent book in another book note, tinyurl.com/LKbooknote.

Still, much though I like this pairing of (chapters from) C&H and L&K, some readers will prefer a shorter presentation of FOL in a single text and in a somewhat more conventional style. For them, I can particularly recommend:

4. Dag Westerståhl, *Foundations of Logic: Completeness, Incompleteness, Computability* (CSLI 2024), Chapters 1 to 5.

This is, by some margin, the most attractively written book at its level that I have newly encountered for some years. It is based on lecture courses given at Stockholm University (and latterly in Beijing), and evidently reflects long teaching experience. Especially in these early chapters, the expository choices are very well-judged, and the balance between motivational chat and worked-through formal details seems just right to me. Many student readers should find this book quite excellent for self-study.

An inviting, reader-friendly, Introduction (Ch. 1), is followed by a substantial presentation (Ch. 2) explaining the syntax and semantics of propositional logic and then of FOL. This is done conventionally, in that we get the usual Tarski-style semantics, assigning objects to all variables and interpreting wffs with free variables.. This is expounded particularly clearly, with motivations at various points all very well explained.

Then Ch. 3 introduces two proof systems, Gentzen-style natural deduction and a minimalist, two-connective, Hilbert-style system. I think I might have spent just two or three more pages giving more examples of Gentzen-style proofs, showing tactics for building up proof-trees in natural ways (to reinforce

the advertised claim that this really is a *natural* deduction system). But still, this is again all very well done, with possible stumbling blocks – for example, the use of vacuous discharge – nicely elucidated. The chapter finishes particularly helpfully with an outline proof that the two proof systems march in step, warranting the same deductions.

There follows a short chapter (Ch. 4) proving the completeness of the Hilbert proof system for propositional logic, using maximal consistent sets and Lindenbaum’s Lemma. As in previous (and future!) chapters, there are excellent exercises, including in this case ones exploring another completeness proof and presenting the interpolation lemma for PL. Then, closely following that chapter’s overall structure, we get a brisk chapter (Ch.5) proving the completeness theorem for FOL using Henkin’s method – very sensibly doing the proof for FOL without identity first, before complicating the story in the final section of the chapter to handle identity. These chapters together give us an excellent and very approachable book-within-a-book, Westerståhl’s introduction to FOL in little over 100 pages. (The following longer Ch. 6 is titled, simply, ‘Model theory’: you could optionally also tackle the earlier sections at this stage. For more about the book, see tinyurl.com/Westerstahl.)

4.2 Some parallel and slightly more advanced reading

(a) FOL is so very fundamental, and the alternative options presenting the same material are so very many, that I really do need to say at least something about other books. So in this section I list – in rough order of difficulty/sophistication – a small handful of further texts which could very well make particularly useful additional or alternative reading (there’s a much longer list of books, mostly treated much more briefly, in §4.4).

First, though, let me mention an excellent text pitched a notch or two down in level from the texts we have looked at so far. This is a book written by a philosopher for philosophers: so it should be particularly accessible to non-mathematicians who haven’t done much formal logic before, and it could very well help ease the transition to coping with the more mathematical style of the books recommended in the last section.

5. David Bostock, *Intermediate Logic* (OUP 1997).

From the preface: “[This] book is confined to . . . what is called first-order predicate logic, but it aims to treat this subject in very much more detail than a standard introductory text. In particular, whereas an introductory text will pursue just one style of semantics, just one method of proof, and so on, this book aims to create a wider and a deeper understanding by showing how several

alternative approaches are possible, and by introducing comparisons between them.”

So Bostock ranges more widely than the books I’ve so far mentioned; he usefully introduces you to semantic tableaux *and* a Hilbert-style axiomatic proof system *and* natural deduction *and* even a sequent calculus as well. Indeed, though written for non-mathematicians, anyone could profit from at least a quick browse of his Part II to get a quick sense of the various approaches.

Bostock eventually touches on issues of philosophical interest such as free logic which are not often dealt with in other books at this level. Still, the discussions mostly remain at much the same level of conceptual/mathematical difficulty as in, for example, my own introductory book. And, to repeat, unlike our main recommendations, Bostock does give a brisk but very clear presentation of tableaux (‘truth trees’), and he proves completeness for tableaux in particular, which I always think makes the needed construction seem particularly natural. (If you are a philosopher, you may well have already encountered truth trees in your introductory logic course. If not, at some point you will want to find out about them.¹)

(b) Next, let me mention two well-regarded and often-used texts. First, a very useful alternative to C&H. Later chapters of this book are also mentioned later in this Guide as providing possible reading on further topics, so it could be well worth making early acquaintance with

6. Dirk van Dalen, *Logic and Structure* (Springer, 1980; 5th edition 2012).

The chapters up to and including §3.2 provide a nice introduction to FOL via Gentzen-style natural deduction. The treatment *is* approachable and the book is written with a relatively light touch.

True, it has to be said that the book isn’t without its quirks and flaws and inconsistencies of presentation: but you would have to be a rather alert and picky reader to notice and be bothered by them. Otherwise, the coverage and general approach is attractive, and mathematicians should be able to cope very readily (and may prefer van Dalen to C&H). I suspect, however, that the book would occasionally be tougher going for philosophers if taken from a standing start – which is one reason why I suggested beginning with C&H instead. For more on this book, see tinyurl.com/dalenlogic.

As a complement to C&H’s natural deduction text, I recommended L&K’s *Friendly Introduction* which uses an axiomatic system. As an alternative to that, here is an older (and, in its day, much-used) book:

¹ As an alternative to Bostock, my elementary account of truth trees for propositional logic available at tinyurl.com/proptruthtrees will quickly give you the basic idea in an accessible way. Then you can dip into my introduction to truth trees for quantified logic at tinyurl.com/qtruthtrees.

7. Herbert Enderton, *A Mathematical Introduction to Logic* (Academic Press 1972, 2002).

This also focuses on an axiomatic system, and is regarded as a classic of exposition. However, it does strike me as somewhat less approachable than L&K, so I'm not surprised that some students do find this book somewhat challenging if used by itself as a first text.

However, this is an entirely admirable and very reliable piece of work which many readers should be able to cope with, certainly if used as a supplementary second text. And stronger mathematicians might even find it their first preference for a text based on an axiomatic system.

Read up to and including §2.5 or §2.6 at this stage. Later, you can finish the rest of that chapter to take you a bit further into model theory. For more about this classic, see tinyurl.com/enderlogicnote.

- (c) Although it is giving a second bite to an author we've already met, let me now mention a rather different discussion of FOL, in an essay aimed at philosophically minded logicians:

8. Wilfrid Hodges, 'Elementary predicate logic', in the *Handbook of Philosophical Logic*, Vol. 1, ed. D. Gabbay and F. Guenther (Kluwer, 2nd edition 2001).

This is a slightly expanded version of the entry in the first edition of the *Handbook* (read that earlier version if this one isn't available), and is written with Hodges's usual enviable clarity and verve. It is full of conceptual insights, historical asides, and comparisons of different ways of doing things – it very nicely complements the textbook presentations mentioned so far. At this stage, read the very illuminating first twenty short sections.

- (d) I'll finish this section by recommending two further books that push the story on a little. First, an absolute classic, short but packed with good things:

9. Raymond Smullyan, *First-Order Logic** (Springer 1968, Dover Publications 1995).

This is terse, but those with a taste for mathematical elegance can certainly try its Parts I and II, just a hundred pages, after the initial recommended reading in the previous section. This beautiful little book is the source and inspiration of many modern treatments of logic based on tree/tableau systems. Not always easy, especially as the book progresses, but it could be a real delight for the mathematically minded.

And second, taking things in a new direction, don't be put off by the title of this excellent book:

10. Melvin Fitting, *First-Order Logic and Automated Theorem Proving* (Springer, 1990, 2nd edn. 1996).

This is a wonderfully lucid book by a renowned expositor.²

As anyone who has tried to work inside an axiomatic system knows, proof-discovery for such systems is often hard. Which axiom schema should we instantiate with which wffs at any given stage of a proof? Natural deduction systems are nicer. But since we can, in effect, make any new temporary assumption we like at any stage in a proof, again we need to keep our wits about us if we are to avoid going off on useless diversions. By contrast, tableau proofs (a.k.a. tree proofs) can pretty much write themselves even for quite complex FOL arguments, which is why I used to introduce formal proofs to students that way (in teaching tableaux, we can largely separate the business of getting across the idea of formality from the task of teaching heuristics of proof-discovery). And because tableau proofs very often write themselves, they are also good for automated theorem proving. Fitting explores both the tableau method and the related so-called resolution method which we mentioned as, yes, a fifth style of proof.

This book's approach, then, is rather different from the other texts I have recommended. However, I do think that the fresh light thrown on first-order logic makes the slight detour through this extremely clearly written book *vaut le voyage*, as the Michelin guides say.³

4.3 Just a little history

An obvious starting place from which to explore something of the history of FOL (and you are warmly encouraged to read at least this!) is the very clear and judicious encyclopaedia entry

11. William Ewald, 'The emergence of first-order logic', *The Stanford Encyclopedia*, tinyurl.com/emergenceFOL.

If you want rather more, the following by another excellent author is also readable and very helpful:

² Yes, at a number of places in the text there are illustrations of how to implement algorithms in Prolog. But either you can easily pick up the very small amount of background knowledge that's needed to follow everything that is going on (and that's quite fun) or you can in fact just skip lightly over those implementation episodes while still getting the principal logical content of the book.

³ If you don't want to take the full tour, however, there's a nice introduction to proofs by resolution in Shawn Hedman, *A First Course in Logic* (OUP, 2004): §1.8, §§3.4–3.5.)

12. José Ferreirós, ‘The road to modern logic – an interpretation’, *Bulletin of Symbolic Logic* 7 (2001): 441–484, tinyurl.com/roadtologic.

For a longer, though rather bumpier, read – you’ll probably need to skim and skip! – enthusiasts could also try dipping into this fascinating and more wide-ranging piece:

13. Paolo Mancosu, Richard Zach and Calixto Badesa, ‘The development of mathematical logic from Russell to Tarski: 1900–1935’ in Leila Haaparanta, ed., *The History of Modern Logic* (OUP, 2009, pp. 318–471): tinyurl.com/devlogic.

4.4 Postscript: Other treatments?

I will end this chapter by responding – often very brusquely – to a variety of frequently asked questions raised in response to earlier versions of the Guide (questions of the form “But why haven’t you recommended X?”). So, in what follows,

- (a) I first consider four deservedly classic mainstream mathematical logic books (now more than fifty years old).
 - (b) I then quickly mention a handful of logic books primarily aimed at philosophers (but only one will be of interest to us at this point).
 - (c) Next I look at nine more recent mathematical logic texts (I again highlight one in particular).
 - (d) Finally, for light relief, I look at some fun extras from an author whom we have already met.
- (a) We start, then, with four classic books, listed in order of first publication. All of them are worth visiting sometime, even if they are not now the first choices for beginners.
- (a1) Elliott Mendelson, *Introduction to Mathematical Logic* (van Nostrand 1964; Chapman and Hall/CRC, 6th edn. 2015).
 - (a2) Joseph R. Shoenfield, *Mathematical Logic* (Addison Wesley, 1967).
 - (a3) Stephen C. Kleene, *Mathematical Logic* (John Wiley 1967; Dover Publications 2002).
 - (a4) Geoffrey Hunter, *Metalogic: An Introduction to the Metatheory of Standard First Order Logic* (Macmillan 1971; University of California Press 1992).

Perhaps the most frequent question I used to get asked in response to early versions of the Guide was ‘But what about Mendelson, Chs 1 and 2’? And indeed, (a1) was, I think, the first modern textbook of its type (so immense credit to Mendelson for that). I no doubt owe my whole career to it – it got me through tripos when the world was a lot younger!

It seems that some others who learnt using the book are in their turn still using it to

teach from. But let's not get too sentimental! It has to be said that the book in its first incarnation was often brisk to the point of unfriendliness; and the basic look-and-feel of the book hasn't changed a great deal as it has run through successive editions. Mendelson's presentation of axiomatic systems of logic is quite tough going (and as the book progresses in later chapters through formal number theory and set theory, things if anything get somewhat less reader-friendly). Which certainly *doesn't* mean the book won't repay working through. But quite unsurprisingly, over fifty years on, there are many rather more accessible and more amiable alternatives for *beginning* serious logic. So, although Mendelson's book is a landmark well worth visiting one day, I can't recommend starting here (especially for self-study).

For a little more on the book, see tinyurl.com/mendelsonlogic.

Shoenfield's (a2) is really aimed at graduate mathematicians, and is not very reader-friendly. Maybe take a look one day, particularly at the final chapter on set theory – but certainly not yet! For a little more, see tinyurl.com/schoenlogic.

Kleene's (a3) – not to be confused with the same author's hugely influential earlier *Introduction to Metamathematics* – goes much more gently than Mendelson: it takes almost twice as long to cover propositional and predicate logic, so Kleene has much more room for helpful discursive explanations. This was in its time a rightly much admired text, and would still make excellent supplementary reading.

But if you *do* want an old-school introduction from the same era, you might most enjoy the somewhat less renowned book by Hunter, (a4). This is not as comprehensive as Mendelson: but it was an exceptionally good textbook from a time when there were few to choose from. Read Parts One to Three on FOL. And if you are finding it rewarding reading, then do eventually finish the book: it goes on to consider formal arithmetic and proves the undecidability of first-order logic, topics we consider in Chapter 6.

Unfortunately, the typography – from pre- $\text{\LaTeX}$ days – isn't very pretty to look at. But try to overlook that: in fact the treatment of an axiomatic system of logic is extremely clear and accessible.

(b) The following seven books are very varied in style, level and content, but are all designed with philosophers particularly in mind.

(b1) Richard Jeffrey, *Formal Logic: Its Scope and Limits* (McGraw Hill 1967, 2nd edn. 1981).

(b2) Neil Tennant, *Natural Logic** (Edinburgh UP 1978, 1990). Now out of print, but downloadable as tinyurl.com/tennantNL.

(b3) Merrie Bergmann, James Moor and Jack Nelson, *The Logic Book* (McGraw Hill 1980; 6th edn. 2013).

(b4) John L. Bell, David DeVidi and Graham Solomon, *Logical Options: An Introduction to Classical and Alternative Logics* (Broadview Press 2001).

(b5) Theodore Sider, *Logic for Philosophy** (OUP, 2010).

(b6) Jan von Plato, *Elements of Logical Reasoning** (CUP, 2014).

(b7) Richard Zach and others, *Sets, Logic, Computation** (Open Logic Project, 2025)

I will return to Jeffrey’s lovely book in a moment, so let’s first look at Tennant’s (b2). All credit to the author for writing the first textbook aimed at an introductory level which does Gentzen-style natural deduction. Tennant thinks that this approach to logic is philosophically highly significant, and in various ways this shows through in his textbook.

Although not as conventionally mathematical in look-and-feel as some alternatives, it is in fact *very* careful about important details. This is not always an easy read, however, despite its being intended as a first logic text for philosophers. However the book is there to freely sample, and some may well find it highly illuminating parallel reading on natural deduction.

The Logic Book (b3) is over 550 pages, starting at about the level of my introductory book, and going as far as results like a full completeness proof for FOL, so its coverage overlaps considerably with the main recommendations of §4.1. But while reliable enough, it all strikes me, like some students who have commented, as *very* dull and laboured, and often rather unnecessarily hard going. You can certainly do better.

Only the first two chapters of *Logical Options* (b4) are on FOL, and not at the level we really want.

Sider’s book (b5) falls into two halves, and the second half is actually quite good on modal logic; but the first half of the book, the part which is relevant to us now, is very poor: you can safely ignore it.

By contrast, Von Plato’s *Elements* (b6) is a very good book. However, though the author would disagree, it is perhaps better regarded as an introduction to proof theory rather than a conventional introduction to FOL and we will return to it in Chapter 10.

The Open Logic Project provides an “open-source, modular, collaboratively authored collection of teaching materials for formal (meta)logic and formal methods . . . aimed at a non-mathematical audience (in particular, students of philosophy and computer science).”

Various texts have been “mixed” from (versions of) chapters available on the project site. In particular, one is “remixed” by Richard Zach under the title *Sets, Logic, Computation: An Open Introduction to Metalogic*, and can be downloaded from slc.openlogicproject.org. Part II provides an introduction to FOL (b7).

Briefly, after a useful enough introductory overview, Chapters 6 and 7 are on the syntax and semantics of FOL. The syntax chapter should be pretty accessible; but the semantics chapter is rather harder going, not entirely helped by some unnecessarily

heavy symbolism. Chapter 8, ‘Theories and their models’, is shorter, less dense, and nicely done.

There are then three chapters on what the book calls “Derivation Systems”. Chapter 9 gives a very snappy overview of a familiar quartet, old-school axiomatic systems, tableaux systems as often taught in first logic courses for philosophers, natural deduction, and sequent calculi. Then Chapter 10 develops a Gentzen-style sequent calculus where sequents have the form $\Delta \Rightarrow \Gamma$, with ordered sequences of formulas on both the left and right. Why sequences rather than sets, I don’t know.

Now, I am as appreciative as the next logician of the great mathematical elegance of a two-sided sequent calculus: but why this as the *first* proof system given an extended treatment in what is intended as a relatively gentle book for the non-mathematical? I suspect many a philosophy student might be puzzled.

Chapter 11 then introduces a Gentzen-style natural deduction system. But there are better presentations out there.

Chapters 10 and 11 already give us soundness proofs for the sequent and natural deduction calculi. Completeness (along with compactness and Löwenheim-Skolem) is then the topic of Chapter 12. And the Henkin-style proof is done clearly enough. So various chapters could make useful supplementary/revision material. But these are not my choices for a first introduction to FOL. For a little more, see tinyurl.com/openzach.

So that leaves Richard Jeffrey’s classic (b1). This is relatively short, and the first half on propositional logic is mostly at a very introductory level, which is why I haven’t mentioned it before. But if you know a little about trees for propositional logic, then you could start at Chapter 5 and read the rest of the book with considerable enjoyment and illumination. For this gives a gentle yet elegant introduction to the undecidability of FOL and a particularly nice proof of completeness for trees.

(c) We now turn to a number of more recent texts in mathematical logic that have been suggested as candidates for this Guide. As you will see, the most interesting of them – which almost made the cut for the list of additional readings in §4.2 – is the idiosyncratic book by Kaye.

- (c1) H.-D. Ebbinghaus, J. Flum and W. Thomas, *Mathematical Logic* (Springer, 2nd edn 1994, 3rd edn. 2021).
- (c2) René Cori and Daniel Lascar, *Mathematical Logic, A Course with Exercises: Part I* (OUP, 2000).
- (c3) Shawn Hedman, *A First Course in Logic* (OUP, 2004).
- (c4) Peter Hinman, *Fundamentals of Mathematical Logic* (A. K. Peters, 2005).
- (c5) Wolfgang Rautenberg, *A Concise Introduction to Mathematical Logic* (Springer, 2nd edn. 2006).
- (c6) Richard Kaye, *The Mathematics of Logic* (CUP 2007).
- (c7) Harrie de Swart, *Philosophical and Mathematical Logic* (Springer, 2018)
- (c8) Martin Hils and François Loeser, *A First Journey Through Logic* (AMS Student

Mathematical Library, 2019).

(c9) Jeremy Avigad, *Mathematical Logic and Computation* (CUP, 2022)

(c10) Joseph Mileti, *Modern Mathematical Logic* (CUP 2023)

The text by Ebbinghaus, Flum and Thomas's (c1) is the English translation of a book first published in German in 1978, and appears in a series 'Undergraduate Texts in Mathematics', which indicates the intended level. The book is often warmly praised and is (I believe) quite widely used in Germany. There is a lot of material here, often covered well. But I can't find myself wanting to recommend it as a good place to *start*. The core material on the syntax and semantics of first-order logic in Chs 2 and 3 is presented more accessibly and more elegantly elsewhere. And the treatment of a sequent calculus Ch. 4 strikes me as uninviting, with the authors failing to capture the elegance that using a sequent calculus can bring. For more on this book, see tinyurl.com/EFTbooknote.

Chapters 1 and 3 of Cori and Lascar's (c2) could appeal to the more mathematical reader. Chapter 1 is on semantic aspects of propositional logic, and is done clearly. Also, an unusually good feature of the book is that there are – as with other chapters – interestingly non-trivial exercises, with expansive answers given at the end. Chapter 2, I would say, jumps to a significantly more demanding level, introducing Boolean algebras (and really, you should probably know a bit of algebra and topology to fully appreciate what is going on – we'll return to this in §13.1). Chapter 3 gets back on track with the syntax and semantics of predicate languages, plus a smidgin of model theory too. Not perhaps the place to start for a first introduction to this material, but worth reading. Then Chapter 4, the last in the book, is on proof systems, but probably not so helpful.

Shawn Hedman's (c3) is subtitled 'An Introduction to Model Theory, Proof Theory, Computability and Complexity'. So there is no lack of ambition in the coverage! The treatment of basic FOL is patchy, however. It is pretty clear on semantics, and the book can be recommended to more mathematical readers for its treatment of more advanced model-theoretic topics (see §5.3 in this Guide). But Hedman offers a peculiarly ugly not-so-natural deductive system.

By contrast though – as already noted – he *is* good on so-called resolution proofs. For more about what does and what doesn't work in Hedman's book, see tinyurl.com/hedmanbook.

Peter Hinman's (c4) is a massive 878 pages, and as you'd expect covers a great deal. Hinman is, however, not really focused on deductive systems for logic, which don't make an appearance until over two hundred pages into the book (his concerns are more model-theoretic). And most readers will find this book pretty tough going. This is certainly not, then, the place to start with FOL. However, the first three chapters of the book do contain some supplementary material that could be very interesting once you have got hold of the basics from elsewhere, and could particularly appeal

to mathematicians. For more about what does and what doesn't work in Hinman's book, see tinyurl.com/hinmanbook.

The first three chapters of Wolfgang Rautenberg's *Concise Introduction* (c5) are on FOL and have some nice touches. But I suspect that these hundred pages will be rather *too* concise to serve many readers as an initial introduction. And his preferred formal system is not a 'best buy' either. This book can, however, be recommended as providing good revision material.

Jumping past Kaye's (c6) for the moment, we get to the relevant Chapters 2 and 4 on FOL in de Swart's rather rambling (c7). These chapters are quite attractively written, and have some interest. But there also are a number of presentation choices I'd quibble with, and you can do better.

Next, we can skip past (c8), which just isn't designed to be a conventional mathematical logic text. It does have a fast-track introduction to FOL, but this is done far *too* fast to be of much use to anyone.

Avigad's *Mathematical Logic and Computation* (c9) is an impressive book, full of good things, with excellent presentations of material not readily to be found elsewhere, from which I learnt a lot. In particular, the first seven chapters are a book-within-a book on FOL, but with a distinctively strong proof-theoretic flavour (and uneven levels of difficulty). My advice, then, is to tackle this after you have read some entry-level proof-theory, as you will then get much more out of Avigad's work. So we will return to this. (For more, see tinyurl.com/javigad-mlc.)

Mileti's *Modern Mathematical Logic* (c10) is perhaps mistitled: it is in many ways a very conventional, old-school, book. And some of the expositional choices seem misguided. For example, Chapter 2 is thirty pages on 'Induction and Recursion'. We get a pretty dense treatment of what Mileti calls 'generating systems', three different ways of defining the set of generated whatnots, proofs that these definitions come to the same thing, etc. This is all done in what strikes me as a heavy-handed way which could be pretty off-putting as a way of starting out.

Then Chapter 3, the next fifty pages, is on propositional logic. My main beef here is with the chosen formal proof system. This is advertised as natural deduction, but it is a sequent system, where on the left of sequents we get sequences rather than sets (why?). And although the rules are set out in a way that would naturally invite tree-shaped proofs, they are actually applied to produce linear proofs (why?). Moreover, the chosen rule-set is not happily motivated.

I could continue (and do continue at tinyurl.com/miletiMML). I wouldn't recommend this book as a place to start on a serious study of FOL. Some parts of this book could provide supplementary reading for self-study. But to be frank, I'm not quite sure what the point of Mileti's text is, given all the alternatives.

Finally, Richard Kaye is the author of a particularly attractively written 1991 classic on models of Peano Arithmetic (we will meet this in §13.3). So I had high hopes

for his later *The Mathematics of Logic* (c6). “This book”, he writes, “presents the material usually treated in a first course in logic, but in a way that should appeal to a suspicious mathematician wanting to see some genuine mathematical applications. . . . I do not present the main concepts and goals of first-order logic straight away. Instead, I start by showing what the main mathematical idea of ‘a completeness theorem’ is, with some illustrations that have real mathematical content.” So the reader is taken on a mathematical journey starting with König’s Lemma (I’m not going to explain that here!), and progressing via order relations, Zorn’s Lemma (equivalent to the Axiom of Choice), Boolean algebras, and propositional logic, to completeness and compactness of first-order logic. Does this very unusual route work as an *introduction*? I am not at all convinced. It seems to me that the journey is made too bumpy and the road taken is far too uneven in level for this to be appealing as an early trip through first-order logic. However, if you *already* know a fair amount of this material from more conventional presentations, the different angle of approach in this book linking topics together in new ways could well be very interesting and illuminating.

(d) After all these more or less conventional texts, let me offer you some logic puzzle books for dessert!

I have already strongly recommended Raymond Smullyan’s 1968 *First-Order Logic*. Smullyan went on to write some absolutely classic texts on Gödel’s theorem and on ‘diagonalization’ arguments, which we’ll be mentioning later. But as well as these, he also wrote many ‘puzzle’-based books aimed at a wider audience, including for example the rightly renowned *What is the Name of This Book?** (Dover Publications reprint of 1981 original, 2011) and *The Gödelian Puzzle Book** (Dover Publications, 2013).

Smullyan has also written *Logical Labyrinths* (A. K. Peters, 2009). From the blurb: “This book features a unique approach to the teaching of mathematical logic by putting it in the context of the puzzles and paradoxes of common language and rational thought. It serves as a bridge from the author’s puzzle books to his technical writing in the fascinating field of mathematical logic. Using the logic of lying and truth-telling, the author introduces the readers to informal reasoning preparing them for the formal study of symbolic logic, from propositional logic to first-order logic, . . . The book includes a journey through the amazing labyrinths of infinity, which have stirred the imagination of mankind as much, if not more, than any other subject.”

Smullyan starts, then, with puzzles of this kind. You are visiting an island where there are Knights (truth-tellers) and Knaves (persistent liars). And then, in various scenarios, you have to work out what’s true given what the inhabitants say about each other and the world. And, without too many big leaps, he ends with first-order logic (using tableaux), completeness, compactness and more. To be sure, this is no substitute for standard texts: but – for those with a taste for being led up to the serious stuff via sequences of puzzles – this is a very intriguing supplement.

4 *First-order logic – what to read?*

(Smullyan's later *A Beginner's Guide to Mathematical Logic**, Dover Publications, 2014, is rather more conventional. The first 170 pages are relevant to FOL. A rather uneven read, it seems to me; but again an engaging supplement to the main texts recommended above.)